Reinforcement Learning for Large-Scale Auto-Bidding Challenge

COS 435 / ECE 433 — Course Project

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Motivation & Background

Setting: An auto-bidding agent bids on ad slots on behalf of an advertiser across $T = 48$ decision steps per delivery period.

Platform: AuctionNet benchmark (NeurIPS 2024). 48 advertisers, 48 timesteps/period, logged data from diverse agents.

At each step t, the agent:

1. Observes impression opportunities with conversion probabilities ($\text{pValue}_i, \text{pSigma}_i$)
2. Chooses a bid scaling factor α_t
3. Bids $\alpha_t \times \text{pValue}_i$ for each slot i
4. Competes in a Generalized Second-Price auction

Two hard constraints:

- **Budget:** total cost $\leq B$
- **CPA:** $\dfrac{\text{total cost}}{\text{total conversions}} \leq \text{CPA}_{\text{max}} = c$

NeurIPS scoring:

$$\text{Score} = R \times \begin{cases} 1 & \text{if CPA} \leq c \\ \left(\frac{c}{\text{CPA}}\right)^2 & \text{if CPA} > c \end{cases}$$

Problem Formulation & Baseline

State: 16-dim observation (AuctionNet default)

Time/budget fractions, avg. bid & winning cost, conversion rates, win rates, volume statistics (each as all-history & last-3 variants).

Action: $\alpha_t \in \mathbb{R}_{\geq 0}$ (bid $= \alpha_t \times$ pValue$_i$)

Key issue: Not fully capture the history $\Rightarrow$ non-Markovian.

Baseline: PPO (two-stage)

1. BC warm-start (5 epochs)
2. Online PPO via simulator (10M steps)

Online training on period 7-26.

Eval: fixed policy, period 27 (held-out)

Result (16-dim): score $\approx$ 24–25

Plateaus with high variance.

[PPO baseline training curve — 16-dim state, 10M steps]

Extension: Observation Augmentation (16 → 60 dim)

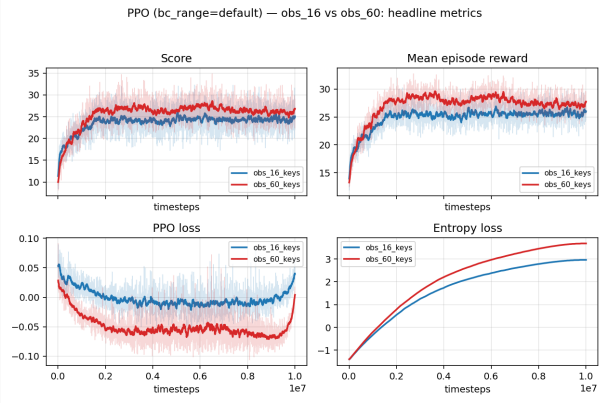
OIL-inspired feature engineering:

Augment state with CPA constraint, budget, per-step cost/conversion history, market signals and etc.

Why it works:

- ✓ Policy conditions on CPA target
- ✓ Captures history ⇒ POMDP → approx. MDP
- ✓ Enables constraint-aware pacing

Result: Higher score, more stable training vs. 16-dim baseline.



Extensions (I)

1. Budget-aware pacing + action gating

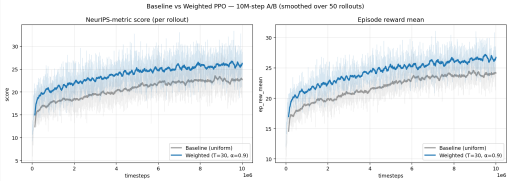
Gate action by remaining budget fraction to prevent overspend.

[Ray's training curve]

2. Replay reweighting

Reweight advertiser sampling with a softmax over each advertiser's historical NeurIPS score, instead of uniform

$w_j = \alpha \cdot \mathrm{softmax}(\mathrm{score}_j / T) + (1-\alpha) \cdot \mathrm{uniform}$



3. Sequence encoder (GRU)

Encode full history into h_t via neural net, replacing "last-3" features.

[Andrew's training curve]

4. Market-regime conditioning

Condition policy on inferred market competitiveness.

[Bonnie's training curve]

Extensions (II) & Final Evaluation

5. Reward shaping

Potential-based: $r' = r + \gamma\Phi(s') - \Phi(s)$

Direct penalty: $r'_t = r_t - \lambda \max(0, \mathrm{CPA}_t - c)$

[Ophelia's training curve]

Final Evaluation (Period 27)

Method	Score	CPA Viol.
PPO baseline (16-d)	$\sim$23	–
+ 60-dim OIL	–	–
+ Budget-aware pacing	–	–
+ Replay reweight	–	–
+ GRU encoder	–	–
+ Market-aware	–	–
+ Reward shaping	–	–
Best combination	–	–

LLM Fine-Tuning: GRPO + LoRA on Qwen3-8B

Setup: Fine-tune Qwen3-8B with LoRA using GRPO.

Training hierarchy:

- 1 trajectory = 1 (advertiser, period), 48 ticks
- 1 rollout row $\rightarrow$ `rollout.n`=4 independent trajectories (same seed, different samples)
- 1 step = `batch_size` $\times$ 4 = 16 episodes

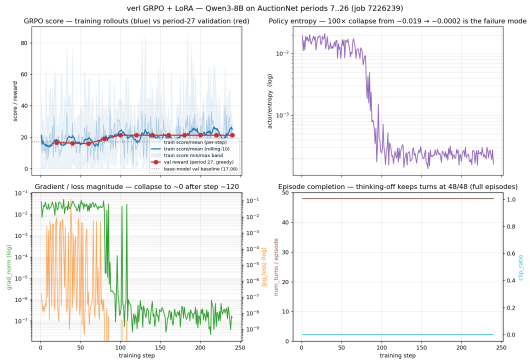
GRPO: Group of 4 rollouts acts as critic.

$A_j = (\text{score}_j - \mu_{\text{group}}) \,/\, \sigma_{\text{group}}$

No separate value network needed.

Key observations:

- Score rises early ($\sim$20 $\rightarrow$ $\sim$30 on train)
- Policy entropy collapses 100$\times$ after step $\sim$120 — mode collapse failure
- Gradients $\rightarrow$ 0, learning stops



Future Directions

Methodological: Combine extensions (e.g., 60-dim + GRU + reward shaping)